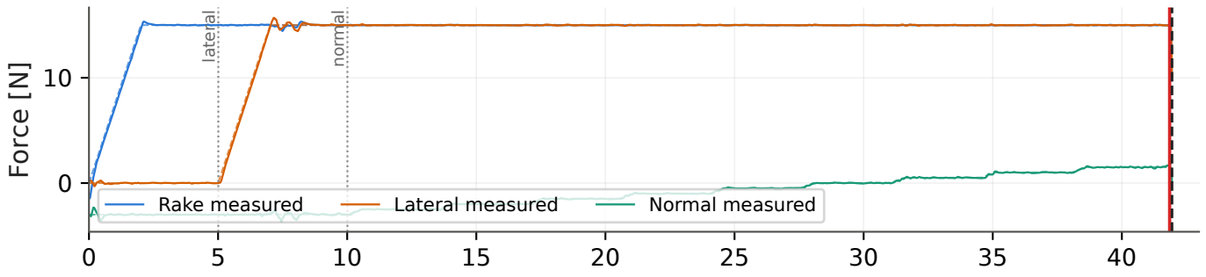
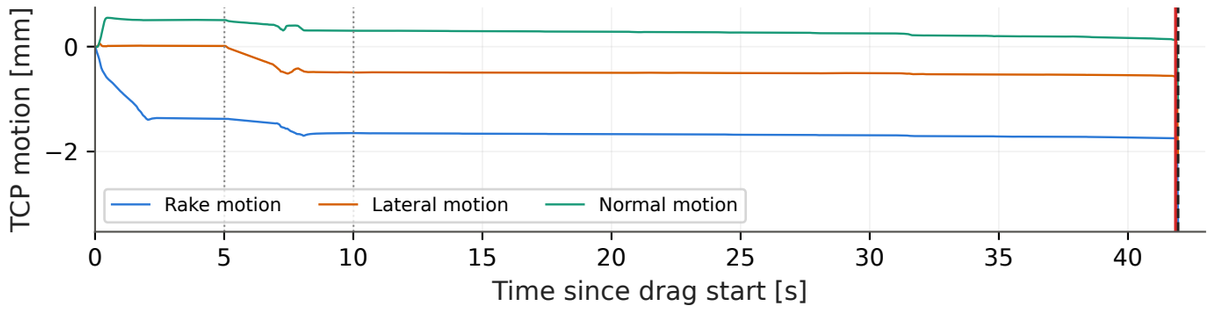
# First slip, re-engagement and final robot stop

Solid = measured; dashed colored = commanded. Red = first event; pale red = later events; black dashed = controller stop.
Motion is reconstructed from TCP pose in a fixed initial frame, avoiding the logged displacement reset between stages.

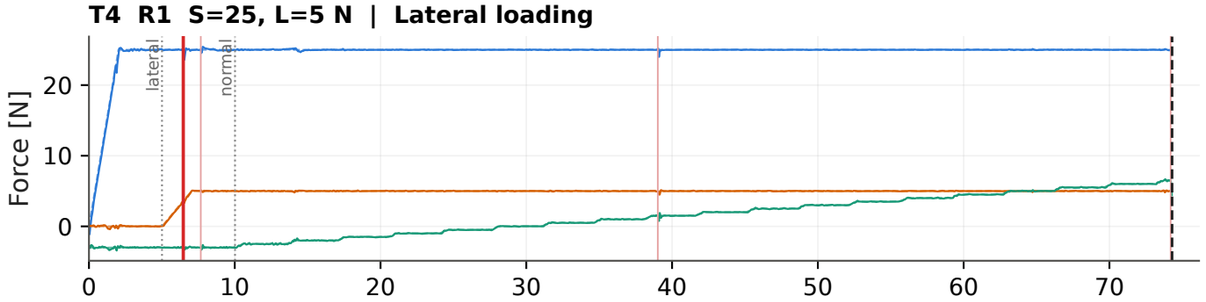
**T1 R1 S=15, L=15 N | Normal tension (Fn > 0)**



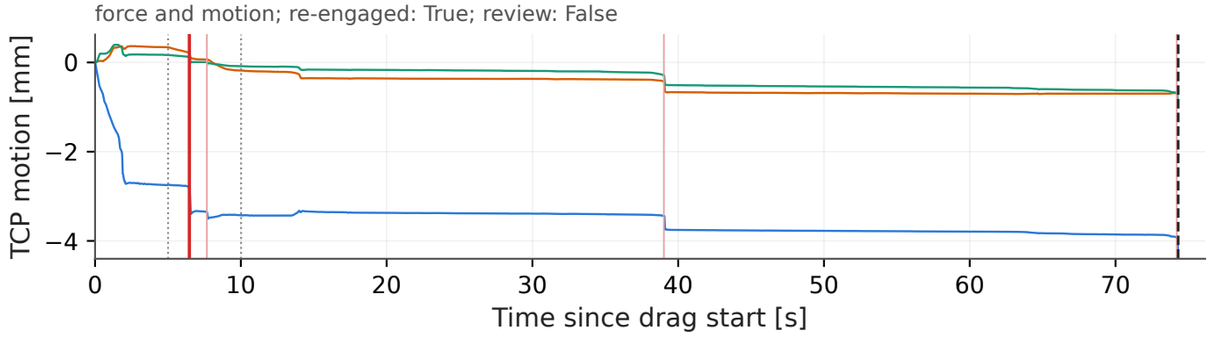

abrupt motion; re-engaged: False; review: False



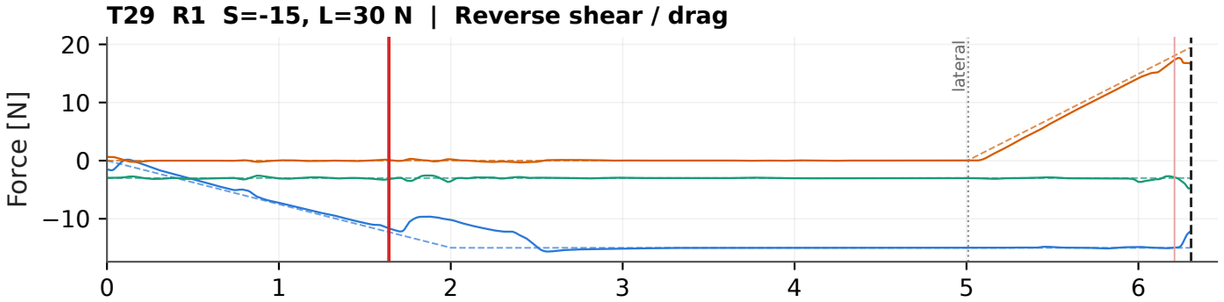

**T4 R1 S=25, L=5 N | Lateral loading**



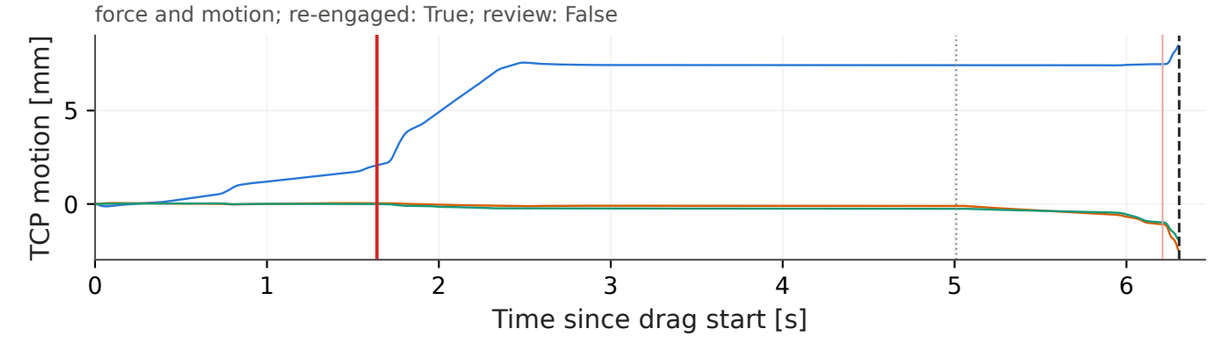

force and motion; re-engaged: True; review: False



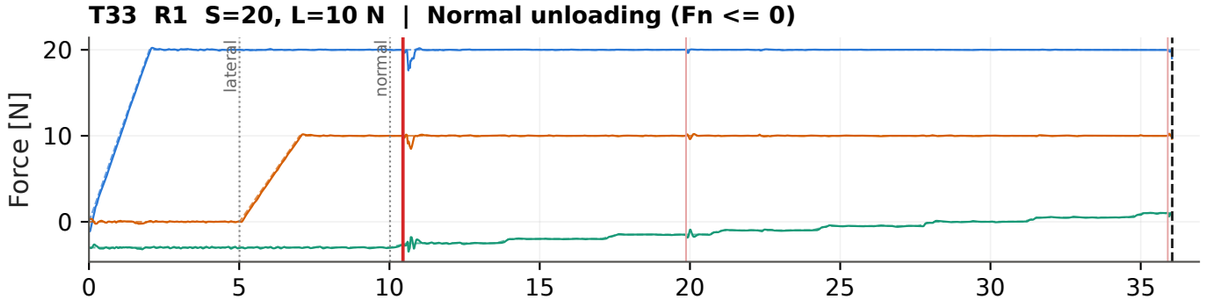

**T29 R1 S=-15, L=30 N | Reverse shear / drag**



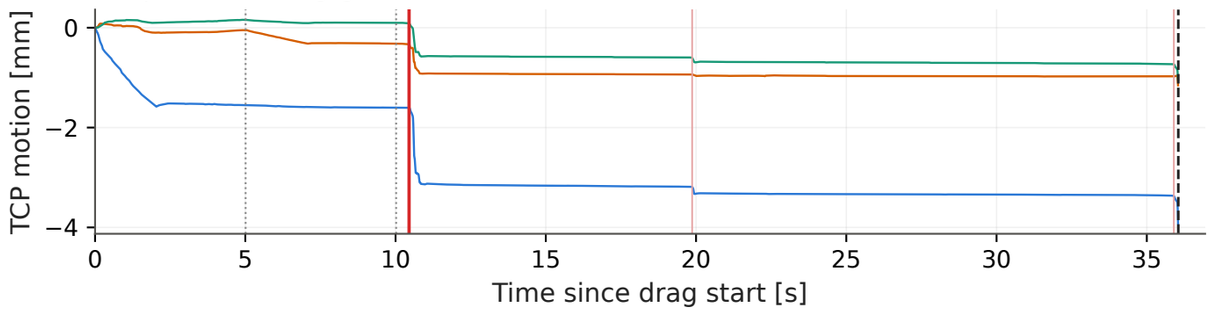

force and motion; re-engaged: True; review: False



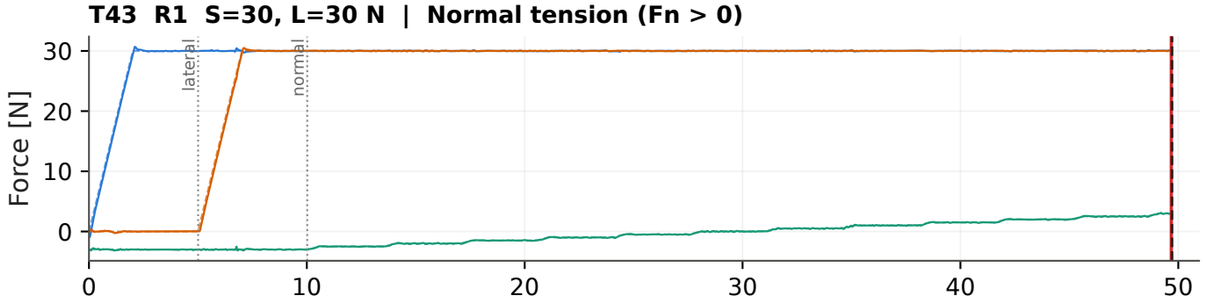

**T33 R1 S=20, L=10 N | Normal unloading (Fn <= 0)**



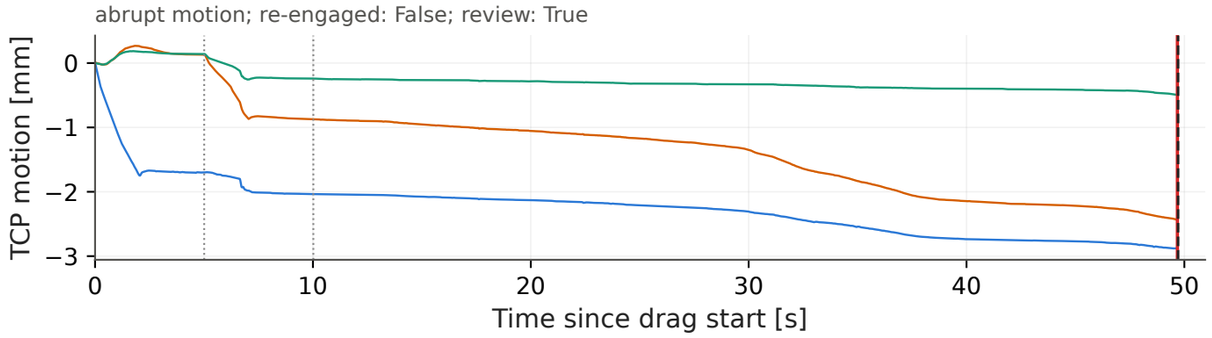

abrupt motion; re-engaged: True; review: False



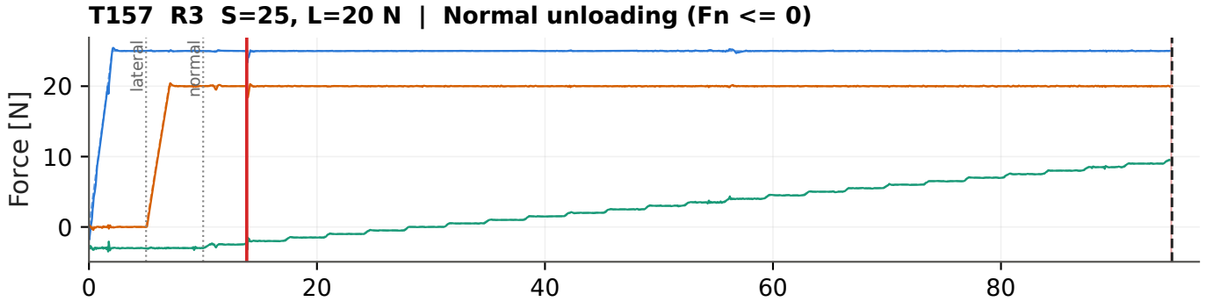

**T43 R1 S=30, L=30 N | Normal tension (Fn > 0)**




abrupt motion; re-engaged: False; review: True




**T157 R3 S=25, L=20 N | Normal unloading (Fn <= 0)**




abrupt motion; re-engaged: True; review: False

